
Bayesian Statistics II: Time Series & Hierarchical Models

BSAD 8310: Business Forecasting — Lecture 11

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Lecture Outline

- 1 Bayesian Time Series: Trend and Seasonality
- 2 Hierarchical Models: Partial Pooling
- 3 Key Takeaways

Bayesian Time Series: Trend and Seasonality

Encode structural knowledge about trend and seasonality directly in the model, with uncertainty estimates on each component.

From Frequentist to Bayesian Time Series

What ARIMA and ETS give you:

- Point forecasts with asymptotic prediction intervals
- Implicit assumptions about trend and error structure
- Limited ability to incorporate domain knowledge
- Single-equation, single-series models

What Bayesian TS adds:

- Full posterior distributions over trend, seasonality, and each forecast horizon
- Explicit priors to incorporate business knowledge ("growth rate is unlikely to exceed 20%")
- Natural uncertainty propagation through time
- Hierarchical structure for multiple related series

Business Application

A retail chain has 500 stores. Rather than fitting 500 independent ARIMA models, a Bayesian hierarchical model shares information across stores:

- Each store gets its own parameters
- Parameters drawn from a common distribution (the hierarchy)
- Sparse stores borrow strength from data-rich stores

Bayesian Structural Time Series

We decompose a series into **trend**, **seasonality**, and **noise**, each with its own priors — giving simultaneous uncertainty estimates on all components.

$$y_t = \mu_t + s_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2)$$

where the trend μ_t evolves as: $\mu_t = \mu_{t-1} + \delta_{t-1} + \eta_t$, $\delta_t = \delta_{t-1} + \zeta_t$
Seasonal component (Fourier features):

Seasonality as smooth sine/cosine waves; more waves (K) = more flexible.

$$s_t = \sum_{k=1}^K \left[a_k \cos\left(\frac{2\pi kt}{P}\right) + b_k \sin\left(\frac{2\pi kt}{P}\right) \right]$$

P = period (12 monthly, 4 quarterly); K controls flexibility.

Local Linear Trend

μ_t : level at time t

δ_t : slope (trend growth rate)

$\eta_t \sim \mathcal{N}(0, \sigma_\eta^2)$: level shock

$\zeta_t \sim \mathcal{N}(0, \sigma_\zeta^2)$: slope shock

Priors on $\sigma_\eta, \sigma_\zeta$ control how much the trend can change over time.

Linear Trend + Seasonality

```
with pm.Model() as ts_model:
    alpha = pm.Normal('alpha',
mu=y.mean(), sigma=y.std())
    beta = pm.Normal('beta', 0, 1)

    t = np.arange(len(y))

    fourier = [pm.Normal(f'a{k}',0,1)*
np.cos(2*np.pi*k*t/12) +
pm.Normal(f'b{k}',0,1)*
np.sin(2*np.pi*k*t/12)
for k in range(1,4)]

    mu = alpha + beta*t + sum(fourier)

    sigma = pm.HalfNormal('sigma', y.std())

    obs = pm.Normal('obs', mu, sigma,
```

What you get:

- Posterior over slope β : growing or declining trend?
- Posterior over a_k, b_k : shape and strength of seasonality
- Posterior predictive: credible bands at each forecast horizon

`pm.sample_posterior_predictive(trace)`
generates forecast draws. Summarize:
median + 90% credible band.

Hierarchical Models: Partial Pooling

Share information across groups via a common prior distribution. The hierarchy learns what is typical; each group learns its own deviation.

Three Approaches to Multi-Group Data

Suppose we have sales data for $J = 50$ stores, each with limited history.

Complete pooling: Fit one model ignoring stores.

- ✗ Assumes all stores are identical
- ✗ Misses systematic store differences
- ✗ Wrong predictions for outlier stores

No pooling: Fit 50 independent models.

- ✗ Ignores shared structure
- ✗ Noisy estimates for stores with few observations
- ✗ Overfits sparse data stores

Partial pooling (hierarchical): Store params drawn from a common prior.

- ✓ Each store has its own estimate
- ✓ Sparse stores shrink toward the group mean
- ✓ Data-rich stores less affected by prior

Partial pooling is the Bayesian answer to the bias-variance trade-off: adding a hierarchical prior reduces variance (less overfitting) at the cost of a small bias (shrinkage toward the mean). (Gelman et al. 2013)

Hierarchical Model: Structure and Code

Partial pooling for store-level intercepts:

$$y_{it} \sim \mathcal{N}(\alpha_i + \beta t, \sigma^2)$$

$$\alpha_i \sim \mathcal{N}(\mu_\alpha, \sigma_\alpha^2), \quad \mu_\alpha \sim \mathcal{N}(\bar{y}, \text{SD}(y)), \quad \sigma_\alpha \sim \text{HN}(1)$$

Key: each store gets α_i , all drawn from $\mathcal{N}(\mu_\alpha, \sigma_\alpha)$. Estimating $\mu_\alpha, \sigma_\alpha$ from data is what makes the model hierarchical.

PyMC Partial Pooling

```
with pm.Model() as hier:
    mu_a = pm.Normal('mu_a', 0, 1)
    sig_a = pm.HalfNormal('sig_a', 1)
    alpha = pm.Normal('alpha',
                      mu=mu_a, sigma=sig_a, shape=J)
    beta = pm.Normal('beta', 0, 1)
    mu = alpha[store_idx] + beta * t
    pm.Normal('obs', mu, 1, observed=y)
```

Non-centered parameterization improves sampling: $\alpha_i = \mu_\alpha + \sigma_\alpha z_i$, $z_i \sim \mathcal{N}(0, 1)$.

Key Takeaways

Bayesian TS and hierarchical models: full uncertainty, shared structure, better predictions for sparse groups.

Lecture 11: Key Takeaways

1. Bayesian time series models decompose the series into **trend**, **seasonality**, and **error**, each with its own prior. The posterior quantifies uncertainty in every component simultaneously.
2. **Fourier features** (cos and sin terms) parameterize seasonality with K harmonics. More harmonics = more flexible seasonal patterns.
3. **Hierarchical models** use partial pooling to share information across groups (stores, regions, products). Sparse groups shrink toward the group mean; data-rich groups are less affected.
4. The three pooling options — complete, no, and partial — represent a bias-variance trade-off. Partial pooling (hierarchical) is almost always the best choice for multi-group business data.
5. In PyMC, hierarchical structure is created by giving group-level parameters a shared prior with hyperparameters that are themselves estimated from the data.

Next: Bayesian Statistics III — Bayesian linear regression, posterior coefficient distributions, and conditional analysis.

References I



Gelman, Andrew et al. (2013). *Bayesian Data Analysis*. 3rd. Boca Raton, FL: Chapman and Hall/CRC.